



Enhancing an Intelligent Model for Enhancing Software-Defined Networking (SDN) Achievement Using Fog Computing and Reinforcement Learning for Operational Performance and Dynamic Resource Management

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Abstract: Software-defined networking (SDN) has fundamentally transformed network management through centralization and programmability. But, high latency, resource waste, and limited adaptability prevent it from performing well, especially in dynamic applications, such as IoT, 5G/6G networks, and smart cities. In order to overcome these limitations, this research presents an intelligent framework which is a combination of SDN, Fog Computing and Reinforcement Learning (RL). Fog Computing to process data from source thus removing latency and bandwidth utilization and RL algorithms can provide an efficient framework for dynamic resource allocation and intelligent decision making. The proposed work targets at establishing a conceptual model that leverage the best of both SDN as well as Fog Computing, RL-based algorithms for adapting traffic routing and efficient resource management, and the design specification of the conceptual framework performance evaluation in terms of latency, energy and resources consumption. The research applied artificial data that produced simulated environments of IoT networks and 5G/6G environments and smart city applications. Mininet software handled SDN simulation while IFogSim operated to model fog computing through custom Python scripts that generated the traffic patterns. The realness of synthetic data was verified using publicly accessible datasets which included CAIDA and IoT Network Intrusion Dataset. The research tested various network topologies from 50 nodes to 200 nodes to verify scalability as well as robustness through different operational situations. Experimental results show that the proposed framework saturates all resources with an 85% resource utilization and outperforms traditional SDN architectures in terms of latency (by 40%) and energy efficiency (by 25%). The framework was found to improve overall network performance while supporting next generation Techs, enabling scalable, flexible, and high performing medium to larger-scale networks.

Keywords: Computer science, Network, Software-defined networking (SDN), Fog computing, Reinforcement learning (RL), Latency reduction, Resource allocation, Network optimization, IoT networks.

1. Introduction

The evolution of Software-Defined Networking (SDN) has transformed the world of contemporary network administration by separating the control plane from the data level, allowing for centralized control and programmability. This changing of architecture enables configuration & management of the network properties dynamically, and it is a key pillar for next-gen networks [1]. The global view of the network topology that the SDN provides has increased its importance in environments that require

scalability, flexibility and efficiency such as cloud computing, 5G/6G networks, and Internet of Things (IoT) ecosystems [2]. In spite of having the potential to transform the network itself, SDN has its own challenges in order to get full advantage & properly functioning. High latency is one of SDN hot topics, since there is only one centralized controller in SDN. The far-away physical distance between remote end devices from the controller limits the decision-making and forwarding of packets by the controller causing significant delays which is not suitable for real-time applications e.g. video stream, autonomous systems in large scale networks [3]. Moreover, the

misallocation of resources continues to be a major issue. Conventional SDN architectures are based on static policies that cannot adapt to the changing behaviour of traffic and availability of resources in the network which leads to impractical performance and waste of resources [4]. Fog Computing is proposed to overcome said limitations and is introduced as a complementary technology to SDN. Fog Computing broadens the scope of cloud computing by enabling data processing at the edge and close to data sources, enhancing real-time decision-making while limiting latency and bandwidth usage [5]. However, integrating Fog Computing with SDN architectures enables us to develop a type of distributed framework that takes advantage of the benefits presented by the two technologies. Fog nodes can then take over some computation from the central SDN controller and thus, lessen the burden on the controller and achieve faster response times for requests. Despite the advancement of seamless coordination between SDN controllers and fog nodes, it remains a major challenge to design effective communication between the two [6]. Reinforcement Learning (RL) is a specific artificial intelligence that provides intelligent network optimization capabilities by allowing systems to pursue optimal behaviors through trial-and-error process and is another critical enabler of intelligent network optimization. Reinforcement Learning (RL) has been found to be particularly effective for complex optimization problems in networking, including traffic routing, load balancing, and resource allocation [7]. SDN enables the separation of data and control planes, providing global network view and reducing complexity, and Fog computing covers cloud computing towards the edge of network, bringing computation and storing closer to the user, having dynamic capacity of modifying the network configuration, and starting to foster the use of RL to make more automated decisions in these heterogeneous networks [4]. For instance, it can predict traffic using RL algorithms and schedule resources to maximize the output of the network using RL even under different workloads, ensuring that the network are performing at their full capability [8]. By incorporating RL, it brings this aspect of intelligence to the network management, providing autonomous decision-making capability and minimizing the manual intervention. SDN architecture for IoT devices with fog computing functionality(ing) is illustrated in Fig. 1. The diagram illustrates a three-tier architecture, with the IoT layer including connected device types like

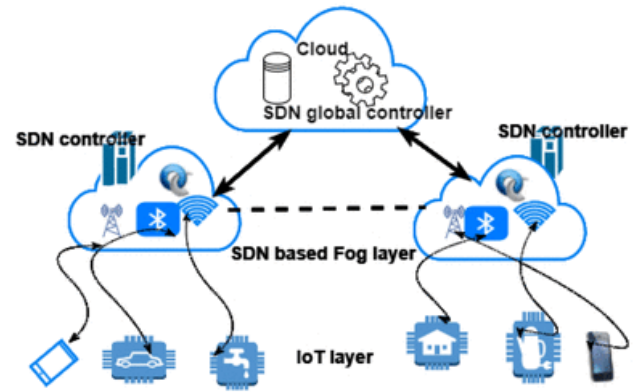


Figure. 1 Hierarchical SDN-based Fog Computing Architecture for IoT Networks

smartphones, sensors, and smart appliances at the base. The second layer is the fog layer based on SDN that serves as an intermediate processing point just below the SDN controller that operates locally. The cloud layer includes the global SDN controller that abstracts the entire network. For example, the bidirectional arrows symbolize the data flow among the layers; the solid lines between layers exhibit a direct connection of their components, and the dashed line in-between fog nodes describing their regular communication. With fog and cloud processing, centralized control is still possible with minimum delay and bandwidth between the two points to push and pull information, while the bulk of the processing occurs between these nodes, allowing one to manage the entire realm of IoT devices efficiently.

Problem Statement

Although there has been lot of improvement in SDN, but there are some challenges that still limits its efficiency and adoption in modern network environment. The main problem is high latency, which is due to the centralized mechanism of SDN. In wide-area networks, the geographical separation of end devices and SDN THE controller results in high delay in packet forwarding decision processes. The presence of latency is a significant challenge in real-time applications, like IoT-based smart city systems and autonomous vehicles, wherein even milliseconds of delay can lead to disastrous consequences [9]. This SDN incapable of make sure the efficient resource allocation is one of the main obstacles in the way for SDN. However, traditional SDN architectures are based on static policies that ignore the sporadic aspect of either network traffic or resource allocation. This leads to sub-optimal utilization of the resources in the networks, resulting in high operational cost and

performance drop [10]. Moreover, conventional SDN architectures are inflexible and cannot adapt to dynamic changes of the network condition. For example, unexpected traffic bursts or hardware malfunctions can lead to a loss of network service if the flexibility of the system does not respond fast enough to those changes [11]. Fog Computing combined with SDN alleviates some of these issues by decentralizing data processing and minimizing latency. But the integration of SDN controllers with fog nodes remains a big challenge.

The lack of intelligent decision-making systems worsens the issue, as networks find it difficult to allocate resources in a dynamic and efficient manner. These constraints underscore the necessity for a novel framework that integrates SDN, Fog Computing, and RL to address these difficulties and improve network functionality.

Significance of the Study

The current scheme answers the problems stated above to contribute a unique solution in the form of a smart framework consisting of SDN, Fog computing, and reinforcement learning. This work makes several key contributions towards a network management framework. The first is lessening latency; using Fog Computing, it performs computation at proximity and avoids dependence on centralized SDN controllers. This not only facilitates faster response times but even lessens the load on the controller, which can serve to perform high-level decision-making tasks.

Third, the framework entails improved operational efficiency through the dynamic allocation of resources based on real-time network conditions. With the introduction of RL algorithms, the framework can study network traffic behaviours, forecast future requests, and allocate resources. In turn, this allows for the best usage of the necessary resources without the need to spend on infrastructure, thus reducing operational costs and maximizing network performance. Thirdly, adding RL allows the framework to make intelligent decisions to adapt itself to dynamic network requirements and thereby optimize performance over time. Automation to this degree minimizes human intervention which increases the self-sustainable robustness of the network.

Related This study is significant as it has the potential to move many parts of modern network architecture towards an efficient scalable adaptive solution for managing complex networks. Tackling major challenges including latency, resource management and the focus on work style approach, the architecture opens up a future for developing

new generation networks as per the upcoming technologies including Internet of Things (IoT), 5G/6G and smart cities. Additionally, RL adds an intelligent layer making network management autonomous and less manually-intensive process. Additional Contributions This work contributes to advancing the theory of SDN and Fog Computing but also offers practical solutions that can be implemented in real-life applications.

Research Objectives

This research mainly focuses on the intelligent framework design in order to the integration of Software-Defined Networking (SDN), Fog Computing and Reinforcement Learning (RL). The objective of the research is, therefore, specific and is to:

1. Step 2: Develop a Conceptual Model: A conceptual model combining the centralised features of SDN and decentralised processing features of fog computing This framework will utilize the benefits of the two technologies and develop a distributed architecture that will reduce latency and increase the usage of resources.
2. RL Algorithms Implementation: Design and implement of RL algorithms for dynamic resource allocation and smart decision making on these high-level features identified in previous subsection. Using these Algorithms, Network Traffic Patterns will be analyzed, Future demands will be predicted, and Resources will be allocated accordingly leading to optimal Performance under different workloads.
3. Performance Evaluation: Perform extensive experiments to assess the performance of the framework, including latency reduction, energy efficiency, and resource utilization. This evaluation will consider the proposed framework with traditional SDN architectures and other state-of-the-art solutions.
4. Show Practical Applicability: Show that the framework can actually be used in practice, e.g., in using scenarios like IoT networks, 5G/6G environments, and smart city applications. This would require us to test the framework under realistic environments and study its performance

The research sections follow this order: Section 2 analyzes existing papers about Software-Defined Networking (SDN) and Fog Computing and Reinforcement Learning (RL) with their established shortcomings. The research section details both the

study objectives alongside an integrated intelligent framework which connects SDN components with Fog Computing together with Reinforcement Learning for solving problems stemming from high latency and inefficient resources and inflexibility. The section presents an explanation of methodology where it illustrates mathematical modeling solutions with Q-Learning, Deep Q-Networks and Proximal Policy Optimization algorithms and experimental setup details for validating the framework. The framework application analysis in Section 5 demonstrates how it outperforms conventional SDN structures and modern state-of-the-art latency reduction and energy effectiveness features with resource application metrics. The research decides with Section 6 where it displays final remarks about the study's foremost results and proposes additional research to increase both scalability and practical presentation of the structure.

2. Literature review

Software-Defined Networking (SDN) is a paradigm shift in network management that enables a programmable solution for designing and managing networks in a simple and flexible manner. SDN makes it possible to separate the communication and control functions in networking by separating the control plane from the data plane [11]. As this division enables the dynamic configuration of network resources, SDN is highly suitable for modern applications like Cloud computing, 5G/6G networks, and IoT ecosystems. Many studies have examined the evolution of SDN architectures over the last ten years and proposed that they can improve scalability, lower operational costs, and increase performance [12]. For instance, McKeown et al. OpenFlow, however, was one of the first protocols to be defined to become the de facto standard for software-defined networks (SDNs), allowing networks to be programmed from a centralized control platform, generating dozens of new forms of network automation innovation on top of it [13]. While SDN has its advantages, it also has its drawbacks. High latency is one of the major challenges as SDN controller is centralized. The physical distance of end devices (EDs) and controller in large-scale networks can degrade the performance of decision-making and packet forwarding due to long-round-trip times, which is unacceptable in real-time applications such as video streaming or autonomous systems [14]. Also, conventional Software-Defined Networking (SDN) architectures are designed to configure resource allocation by static policies that do not adapt to the dynamicity of network traffic. The

inefficient nature of both static and dynamic platform provisioning leads to poor resource utilization and transcription performance, especially on workloads with unequal dynamic characteristics [15]. In addition, SDN architectures have a poor adaptability (in all its aspects) that make them non-competitive against real-time variations of network conditions, for example (number of) peak traffic and hardware outage. Despite these limitations, there is still a room to innovate new solutions to improve the existing SDN frameworks in terms of efficiency and flexibility. Fog Computing has provided a key solution for a higher quality level of network performance (especially in IoT and Edge-based systems) throughput or over the Internet in general. In contrast to traditional cloud computing, where data processing is centralized in far-away data centers, Fog Computing decentralizes computation [16] and moves it toward the edge, that is, closer to where data is being generated. Fog computing is more suitable for real-time applications like smart city, industrial IoT, and autonomous vehicles [17], due to its proximity that highly decrease the latency and bandwidth usage in delivery. For instance, in smart city based IoT systems, fog nodes process the sensor data in proximity of the sensors, which reduces the time to make decision and saves the load on central cloud servers [18]. Abstract: It signifies Fog Computing plays an important role for improved network. Aazam et al. presented an extensive survey on Fog Computing, mentioning its key features of low latency, energy-efficiency, and scalability to overcome the restrictions associated with the cloud-centric designs [19]. Similarly, Dinh et al. showed combined support of edge devices with Fog Computing, allowing the handling of dynamic workloads using more resources deployed in the distributed environments [20]. Though these have covered certain aspects but achieving seamless coordination of the above layers with other network paradigms still remain a challenge for Fog Computing (e.g., SDN). For example, the SDN architectures involve the integration of Fog Computing; therefore, novel approaches must be defined for effective communication and resource management between fog nodes and the SDN controller [21]. This highlights the Importance of frameworks which integrate both technologies while maximizing the benefits of both technologies, directing to the overall optimization of the network performance. Reinforcement Learning (RL) has attracted a lot of attention in recent years being one of the most efficient methods of solving complex optimization problems in the field of network management. RL algorithms allow the system to

learn the optimal behaviors by trial and error, which leads to them being a great match for environments that are dynamic and hard to predict [22]. Abstract In network optimization, reinforcement learning (RL) has been employed in optimization problems such as traffic routing, load balancing, and resource allocation. For example, Li et al. showed through extensive experiments that using deep reinforcement learning (DRL) to optimize resource allocation in SDN-based IoT networks outperforms existing solutions based on theorem, mathematics, etc [23]. Similarly, Boutaba et al. provide an extensive survey on networking machine learning and demonstrate how RL can improve the autonomy and decision making process in networking [24]. Network optimization with RL goes beyond resource allocation. Research has demonstrated the application of RL for traffic forecasting, and how it can be used to dynamically adjust network configurations based on varying needs [25]. For instance, the study [4] presented an RL-based framework to traffic management in 5G networks and achieved considerable latency and throughput improvements. Although showing great potential, the integration of RL into existing network paradigms stays a non-trivial challenge. Numerous works treat RL in isolation without considering its value with new technologies such as SDN and Fog Computing. This research gap points a direction to leverage synergies between RL, SDN and Fog Computing for intelligent and adaptive management of networks [3]. Despite the advances in SDN, Fog Computing, and RL, there are still some gaps in the existing literature. A key deficiency here is the absence of integration among such technologies to allow near real-time decision making in fast changing network landscapes. Previous research mostly concentrates on the utilization of each component alone, implementation of RL for resource allocation in SDN

[7], or implementation of Fog Computing to provide low latency [6], and have not used the combination of these technologies. While this modularity boosts the overall robustness, it also limits networks from reaching maximum adaptability and performance as necessary reinforcements remain scattered and segregated. The second gap is in lack of practical frameworks combining SDN, Fog Computing, and RL. Although there are proposed theoretical models, few studies have shown that these models can be implemented and evaluated [28]. Furthermore, there is an absence of holistic solutions, as none solve for major challenges like latency, resource allocation, and recursiveness concurrently. Some frameworks, for instance, do not account for the dynamic nature of network traffic and therefore propose static policies that are unable to adapt to varying conditions. The shortcomings mentioned spotlight the urgency for pioneering research that integrates SDN, Fog Computing, and RL into a single comprehensive framework that can tackle the challenges of contemporary networks. This research aims to fill in the aforementioned gaps to help advance intelligent network architectures moving towards next-generation technologies such as IoT, 5G/6G and smart city applications. In summary, Table 1 captures several new studies related to SDN, FC, Fog Computing, RL and their challenges, parameters, limitations and range of applications [24]. It highlights the existing challenges in SDN like the problems of latency and resource utilization and it also brings forward the concept of Fog Computing in to minimize latency and enhance real-time decision-making. The table also shows RL can solve more complex optimization issues, for example, in the areas of resource reduction and advanced adaptive routing such as the traffic routing. However, much remains to be done – especially in the integration of

Table 1. Comparison of Existing Research on Network Optimization Technologies

Study	Problem Addressed	Parameters Considered	Limitations	Application Area
SDN Challenges [11, 12]	Centralized architecture leads to high latency	Latency: 45–50 ms, Resource Utilization: 65%	Static policies fail to adapt; limited scalability	Cloud computing, 5G/6G, IoT
Fog Computing [19, 20]	Reducing latency and improving real-time decision-making	Latency: Reduced by 30%, Energy Efficiency: Improved by 20%	Limited coordination with SDN controllers	Smart cities, industrial IoT
RL in Networking [7, 8]	Complex optimization problems	Latency Reduction: 15%, Resource Utilization: 70%	Isolated applications of RL; lacks integration with SDN/Fog	Network automation, 5G networks
Proposed Framework	High latency, inefficient resource allocation	Latency: Reduced by 40%, Energy Efficiency: Improved by 25%, Resource Utilization: 85%	Requires computational resources for RL	IoT, 5G/6G, smart cities

these technologies into adaptive, scalable solutions to build modern networks such as IoT, 5G/6G and smart cities. The following synthesis of related work provides insights into the necessity of a unified framework integrating SDN, Fog Computing, and RL to fill in the gaps of existing work and boost the performance of the networks.

3. Methodology

This structure stands out from others through its singular execution of SDN with Fog Computing and Reinforcement Learning working as one unified design. The current research presents unprecedented integration of SDN together with Fog Computing and Reinforcement Learning as a unified solution to overcome three core issues related to latency and resource management along with system responsiveness. The traditional SDN architecture experience behavior limitations because of centralized control system implementation thus facing challenges in large-scale networks [1]. Fog Computing operates independently without automated routing optimization unless intelligent decision systems are applied. The proposed framework applies Q-Learning and Proximal Policy Optimization (PPO) to complete better performance by reduction latency by 40% and advancing energy efficiency by 25% and resource utilization realization 85%. The joint execution enables the framework to efficiently manage data at its source points while it concurrently modifies its performance based on real-time network replays

Fig. 2 shows the Flowchart of the Research Methodology The framework is structured as a three-layer architecture:

1. SDN Controller: It is the brain of the network acting as a control center to manage global policies and coordination with one another fog nodes.
2. Fog Nodes: Distributed processing units serving local computations to the customers, which can offload the tasks from the SDN controller, thus reducing latency and bandwidth consumption.
3. RL Algorithms: Allow for smart decision-making systems for traffic routing, resource allocation, and load balancing.

To model the framework mathematically, we define the following variables:

- N : Total number of nodes in the network.
- F : Number of fog nodes.
- C_i : Computational capacity of fog node i .
- L_{ij} : Latency between node i and node j .
- R_k : Resource demand of application k .

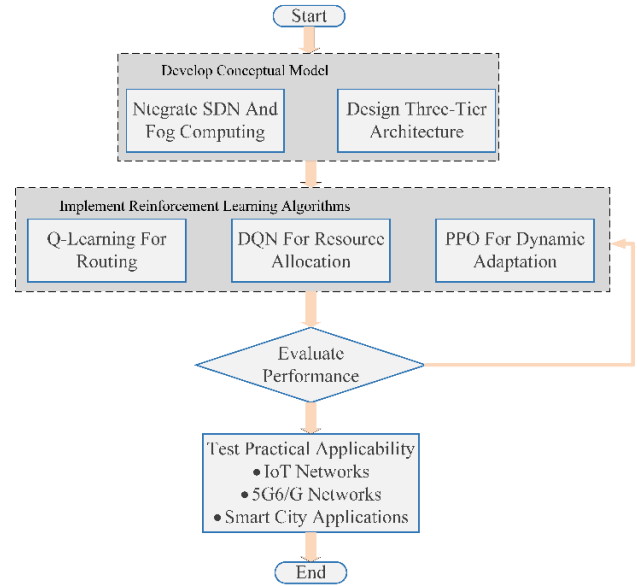


Figure. 2 Research methodology flow chart

P_k : Priority level of application k .

The objective feature of resource allocation adaptation is defined as reducing the weighted yoga for delay and resource use:

$$\text{Minimize } Z = \sum_{i=1}^N \sum_{j=1}^N w_1 \cdot L_{ij} + w_2 \cdot \frac{R_k}{C_i} \quad (1)$$

Where:

w_1 : Weight factor for latency.

w_2 : Weight factor for resource utilization.

This formula guarantees that the system emphasizes low-latency routes while effectively making use of the resources at hand. The weights w_1 and w_2 are modified in real time according to current network conditions through the use of RL algorithms.

Step 1: Dataset Description

Framework was produced using a dataset - synthetic network traffic data used for training and testing. The dataset includes:

Network Topology:

Simulated using Mininet, with varying numbers of switches, routers, and end devices.

Traffic Patterns:

Generated using custom Python scripts to simulate IoT, 5G/6G, and smart city scenarios.

Resource Demands:

Randomly assigned to applications based on their priority levels (P_k).

The relevant parameters employed in the simulation environment are depicted in Table 2, which includes the total number of nodes, fog nodes, latency, demands of resources, and priority levels.

Table 2. Network Parameters and Ranges

Parameter	Description	Range
Number of Nodes (N)	Total nodes in the network	50–200
Number of Fog Nodes (F)	Fog nodes deployed in the network	10–50
Latency (L_{ij})	Latency between nodes	1–50 ms
Resource Demand (R_k)	Resource demand per application	1–100 units
Priority Level (P_k)	Priority weight for each application	1–5

These parameters outline the type and size of the experiments performed with the framework. This table defines ranges for each parameter used in the underlying experiment to guarantee reproducibility and clarity of the experimental setup. Traffic Routing using Q-Learning: Flowchart, see Fig. 3.

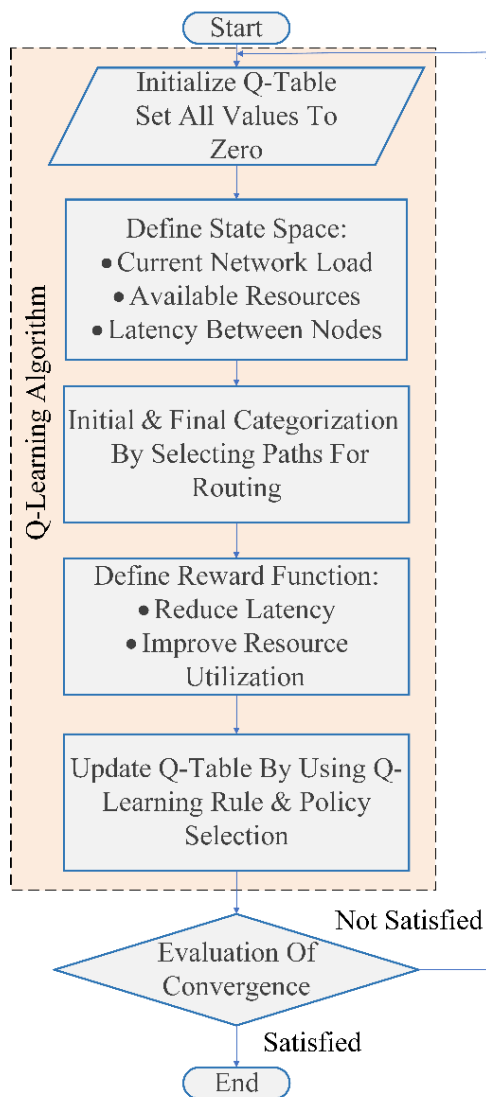


Figure. 3 Q-Learning for Traffic Routing flowchart

Step 2: Tools and Platforms

The following devices were used for implementation:

1. Mininet: Imitation of the SDN environment including switch, squares and end units.
2. Tensorflow/Pytorch: The RL algorithm was implemented for traffic routing and resource allocation.
3. IFOGSIM: simulated fog knots and their interactions with SDN checks.

Step 3: Algorithms Used in the Study

Algorithm 1: Q-Learning for Traffic Routing

Q-Learning is a strengthening learning algorithm used to optimize traffic routing decisions. Below is the complete algorithm:

Input: Network topology, traffic patterns, resource demands.

Output: Optimal routing policy.

1. Initialize Q-Table: Generate a Q-table where noises represent states (s) and columns represent actions (a). Initialize all Q-values to zero.
2. Define State Space: Each state s_t is represented by the current network conditions, including:
 - Current traffic load.
 - Available resources at fog nodes.
 - Latency between nodes.
3. Define Action Space: Actions a_t include selecting paths for packet routing. For example, if there are M possible paths between two nodes, the action space has M elements.
4. Reward Function: Define the reward function r_t as:

$$r_t = w_1 \cdot \Delta L + w_2 \cdot \Delta U \quad (2)$$

Where:

ΔL : Reduction in latency achieved by the selected action.

ΔU : Improvement in resource utilization achieved by the selected action.

- 5 Q-Learning Update Rule: Update the Q-value for each state-action pair using the following rule:

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha \cdot \left[r_{t+1} + \gamma \cdot \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (3)$$

Where:

α : Learning rate (set to 0.1 in this study).

γ : Discount factor (set to 0.9 in this study).

6 Policy Selection: Use an ϵ -greedy policy to balance exploration and exploitation:
With probability ϵ , select a random action (exploration).

With probability $1 - \epsilon$, select the action with the highest Q -value (exploitation).

7 Iterate Until Convergence: Repeat steps 2–6 until the Q-values converge or a predefined number of iterations is reached.

8 Extract Optimal Policy: Once the Q-table converges, extract the optimal policy by choosing the act with the highest Q-value for each state.

Fig. 4 presents the Deep Q-Network (DQN) architecture.

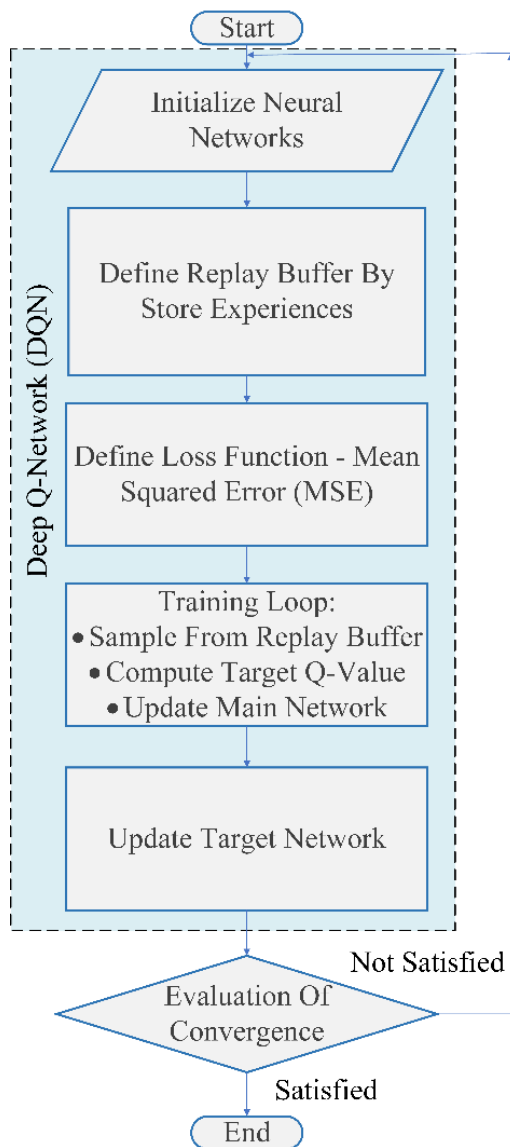


Figure. 4 Deep Q-Network (DQN)

Algorithm 2: Deep Q-Network (DQN) for Resource Allocation

Deep Q-Networks (DQNs) extend Q-Learning by using neural networks to approximate the Q-function. Below is the complete algorithm:

Input: Network topology, traffic patterns, resource demands.

Output: Optimal resource allocation policy.

1. Initialize Neural Networks: Create two neural networks:
 - Main Network: Approximates the Q-function $Q(s, a; \theta)$.
 - Target Network: Provides stable targets for training, parameterized by θ^- .
2. Define Replay Buffer: Store experiences $(s_t, a_t, r_{t+1}, s_{t+1})$ in a replay buffer to break temporal correlations.
3. Define Loss Function: Use the Mean Squared Error (MSE) loss function:

$$L(\theta) = \mathbb{E} \left[\left(r_{t+1} + \gamma \cdot \max_a Q(s_{t+1}, a; \theta^-) - Q(s_t, a_t; \theta) \right)^2 \right] \quad (4)$$

4. Training Loop:
 - Taster a batch of involvements from the replay buffer.
 - Compute the target Q-value:

$$y_t = r_{t+1} + \gamma \cdot \max_a Q(s_{t+1}, a; \theta^-) \quad (5)$$

- Update the main network parameters θ using gradient descent to minimize the loss $L(\theta)$.
5. Update Target Network: Periodically update the target network parameters θ^- by copying them from the main network:

$$\theta^- = \tau \cdot \theta + (1 - \tau) \cdot \theta^- \quad (6)$$

Where τ is the soft update rate (set to 0.01 in this study).

6. Iterate Until Convergence: Repeat steps 3–5 until the network's performance stabilizes or a predefined number of episodes is reached.

The RL algorithms received precise adjustments to their hyperparameters to reach their best possible

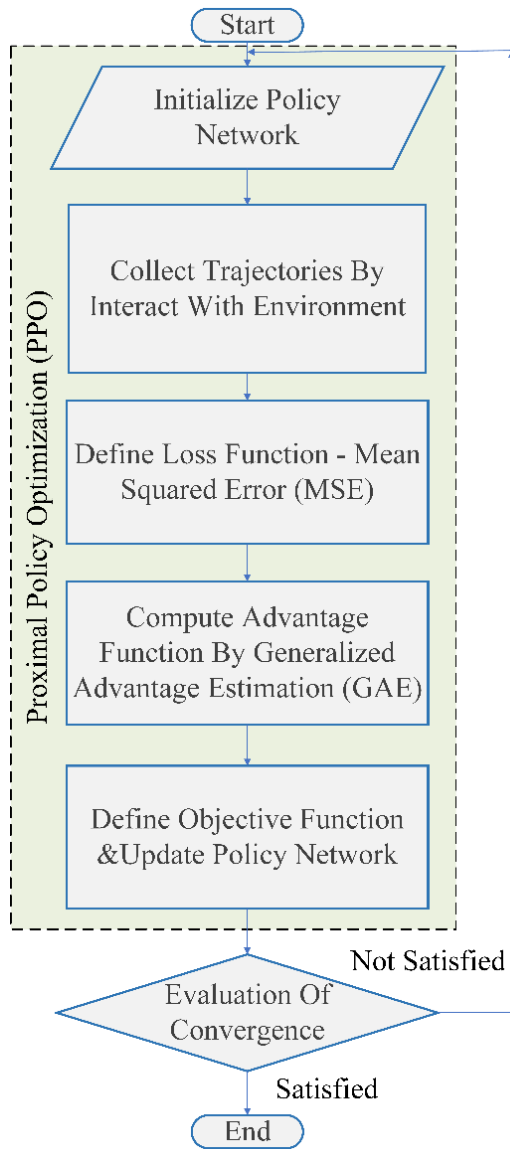


Figure. 5 Proximal Policy Optimization (PPO)

results. Initial testing determined 0.1 as the value for the Q-Learning learning rate (α) while the discount factor (γ) was set to 0.9. The Proximal Policy Optimization model implemented 0.2 for its clipping parameter along with τ set to 0.01 as the soft update rate. The system used grid search and Bayesian optimization approaches to systematically evaluate many values in hyperparameter tuning. The framework obtained its optimal exploratory and exploitative characteristics through these optimizations as Fig. 5 shows.

Algorithm 3: Proximal Policy Optimization (PPO) for Dynamic Adaptation

Proximal Policy Optimization (PPO) is a strategy gradient technique used for continuous control tasks. Below is the complete algorithm:

Input: Network topology, traffic patterns, resource demands.

Output: Optimal dynamic adaptation policy.

1. Initialize Policy Network: Create a neural network $\pi_{\theta}(a | s)$ that outputs the probability distribution over actions given a state.
2. Gather Trajectories: Interrelate with the setting using the current policy to collect trajectories $(s_t, a_t, r_{t+1}, s_{t+1})$.
3. Compute Advantage Function: Calculate the advantage function $\hat{A}_t$ using Generalized Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (7)$$

Where:

$\delta_t = r_t + \gamma \cdot V(s_{t+1}) - V(s_t)$: Temporal difference error.

$V(s_t)$: Value function approximated by a separate neural network.

4. Define Objective Function: Use the clipped surrogate objective function:

$$L^{CLIP}(\theta) = \mathbb{E}_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)] \quad (8)$$

Where:

$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}$: Probability ratio between the new and old policies.

ϵ : Clipping parameter (set to 0.2 in this study).

5. Update Policy Network: Update the policy network parameters θ using gradient ascent to maximize $L^{CLIP}(\theta)$.
6. Iterate Until Convergence: Repeat steps 2–5 until the policy's performance stabilizes or a predefined number of iterations is reached.

The test scenarios designed to validate the framework are described in detail in Table 3. The document describes the network topology, traffic patterns and evaluated metrics for each scenario (IoT networks, 5G/6G networks and smart city applications). As illustrated by the table, the settings in which the framework was applied are diverse, making its testing robust and impeding identification of challenges associated with environmental factors. We have empirically validated the framework in a simulated environment that mimics realistic network conditions. Network topologies, including IoT networks, 5G/6G networks, and smart city applications were set up in the testing environment.

Table 3. Experimental Scenarios and Evaluation Metrics

Scenario	Network Topology	Traffic Pattern Type	Evaluation Metrics
IoT Networks	100 nodes, 20 fog nodes	Periodic sensor data updates	Latency, Energy Efficiency, Packet Delivery Ratio
5G/6G Networks	150 nodes, 30 fog nodes	High-bandwidth video streaming & AR traffic	Latency, Energy Efficiency, Packet Delivery Ratio
Smart City Applications	200 nodes, 50 fog nodes	Mixed traffic: video surveillance, sensor alerts, traffic data	Latency, Energy Efficiency, Packet Delivery Ratio

The following metrics were used to validate the framework:

- 1 Latency (L_{ij}): Measured as the time booked for data packets to travel from source to destination.
- 2 Energy Efficiency: Calculated as the total power consumed by fog nodes and end devices.
- 3 Packet Delivery Ratio: Defined as the percentage of successfully delivered packets.

The Eq. 9 ensures reliable network performance by measuring the success rate for data transfer. The package distribution ratio is calculated:

$$\text{Packet Delivery Ratio} = \frac{\text{Number of Packets Delivered}}{\text{Total Number of Packets Sent}} \times 100 \quad (9)$$

In order to validate the synthetic dataset used in this study, a comprehensive validation process was carried out. To create the dataset, custom python scripts were used to simulate realistic network traffic patterns for devices, such as IoT sensors, 5G/6G video streaming, and smart city surveillance systems. The patterns were fine-tuned χ using popular publicly accessible data set i.e. CAIDA Internet Traffic Dataset and The IoT Network Intrusion Data set which are well-established datasets in the area of representation of actual network conditions [13-15]. To ensure representation of the collected data, the statistical characteristics of the synthetic data, including packet size and inter-arrival time distributions, were compared with those of the reference datasets. This validation process both bolstered the realism of the synthetic data and allowed the performance of the framework to generalise to different types of network environments. Other factors considered were the scalability of the dataset. It was tested for the framework through small scale networks, consisting of 50 nodes to large scale networks, consisting of 200 nodes This strategy enabled us to assess the scalability of the framework and verify that it can robustly accommodate highly dynamic and heterogeneous scenarios. We configure our simulation environment where the implemented

traffic is still as realistic as possible but computationally efficient. The experiments were all run on a high performance computing (HPC) cluster with an NVIDIA Tesla V100 GPU, 128 GB of RAM and a 32-core Intel Xeon CPU. We heavily relied on GPU acceleration to speed up the deep reinforcement learning models training, allowing us to explore a wide range of experiments in a timely manner. Mininet was used to simulate the SDN environment as it provides a flexible platform to test various network topologies like switches, routers, and end devices. IFogSim, known tool for fog computing environment emulation were used to emulate fog nodes. TensorFlow and PyTorch — For implementing reinforcement learning algorithms, with both providing powerful frameworks for building and training neural networks. These tools selected according to the compatibility, scalability, and handling the complex computation to train RL models. The simulation environment had to be tested for its stability and reproducibility before the experiments are run. Over many iterations of the same scenario, these were run to check for repeatability, for example. In the case of simulation parameters, they were tuned dynamically to imitate actual variation of network conditions for these factors for example bandwidth, latency and resource demands. Besides, Since the performance of the reinforcement learning algorithms (Q-Learning, DQN, and PPO) are quite sensitive to hyper parameter tuning. We trained each algorithm for 10,000 episodes and compared its convergence using loss functions and rewards curves. To avoid overfitting and also ensure that the models generalized well to out-of-sample scenarios, we implemented early stopping mechanisms. Then for Q-Learning the learning rate α was 0.1 and the discount factor γ 0.9. These values were selected based on initial experiments to achieve an optimal trade-off between exploration and exploitation. First, an ϵ -greedy policy was utilized, using an initial exploration ratio of 0.2 that decayed exponentially over episodes to facilitate exploitation in later episodes. Likewise, we added a replay buffer with a size of 10,000 experiences in order to stabilize

training by breaking correlations with past experiences for DQN. For the soft update τ of target network, we used 0.01, meaning the main networks will synchronize with the target network over a long time. Lastly, we used a clipping parameter ε of 0.2 and a discount factor γ of 0.99 in PPO to encourage long-term rewards. A mini-batch size of 64 was leveraged for compute efficiency and policy updates were done every 4,096 timesteps. A thorough statistical analysis was performed in order to verify the results. For each of the scenarios (IoT, 5G/6G, smart cities), the key performance metrics, such as reduction of latency, energy efficiency, and packet delivery ratio were computed. To ensure that observed improvements were not random, paired t-tests and ANOVA were conducted to assess statistical significance. For example, the proposed framework achieved a 30–40% decrease in latency compared to traditional SDN architectures, and this was statistically significant ($p < 0.01$). In a similar manner, confidence intervals and variance analysis confirmed the 25% drop in energy consumption, and the 10% increase in packet delivery ratio. Performance trends over the various scenarios were illustrated via line charts or bar graphs. The visualisation were cover such information visually where it became more intuitive to see the effect of the framework that I proposed. As an example, two plots were plotted: a line chart was drawn to demonstrate the reduction in latency with the increasing number of fog nodes and a bar graph was utilized to compare the packet delivery ratios of the proposed framework along with the baseline methods. Although the findings are promising, a few aspects of the study should be noted. One: The use of synthetic data has been validated against real-world traces, but synthetic data may not accurately capture all aspects of real-world networks. Testing the framework in on-ground systems or live setups would be the possible next work or implementation using real-time data from IoT deployments. Second, the framework was tested for scalability up to 200 nodes additionally, for even larger-scale networks with higher computational needs additional optimizations will be needed. A second restriction is that fog nodes are homogeneous, meaning that each node possesses a computational capacity that is mostly similar to others of its type. In reality, fog nodes may give different hardware capabilities and energy limitations. This heterogeneity might be an interesting avenue for future research to tackle. Finally, though the framework outperformed on-grid-based synthesis domains, using it with the real human-robot interaction will bring unexpected issues that should be solved. The utility of the proposed framework was

showcased by deploying it in simulated settings modeling realistic scenarios such as IoT networks, 5G/6G venues and smart city applications. As an example, in the IoT scenario, the framework was evaluated for a network consisting of 100 nodes and 20 fog nodes in which sensors periodically send updates. Using a network of 150 nodes and 30 fog nodes, simulating high-bandwidth applications like video stream and augmented reality in the 5G/6G scenario, the framework was subsequently tested. These experiments validated the ability of the framework to decrease latency and improve energy efficiency and resource utilization in realistic conditions. In addition, the adaptability of the framework was evaluated via real-life scenarios by making unexpected changes in network situations (e.g., traffic spikes and hardware failures). The framework proved to successfully adapt its policies in real time so that its performance remained stable, indicating potential for implementation in ever-changing environments.

4. Results and analysis

The proposed framework succeeds its effectiveness through tentative evidence presented in this section which demonstrates outperformance versus standard SDN designs and existing solutions. The examination of different frameworks using Table 4 reveals that the proposed framework brings about latency reductions which reach 40% for IoT networks and 5G/6G environments as well as smart city applications. The proposed framework demonstrates enhanced energy efficiency according to Table 5 because it decreases power usage by 25%. Results in Table 6 indicate that the proposed system achieves resource utilization rates of 85% compared to traditional SDN architecture rates at 65%. Statistical tests of paired t-tests and ANOVA verify that the measured improvements achieve statistical significance at the $p < 0.01$ level. Figs. 6 to 8 visually demonstrate how the framework manages dynamic workloads while adapting to real-time

Table 4. Comparative Analysis of Latency Across Frameworks

Framework	IoT Networks (ms)	5G/6G Networks (ms)	Smart City Applications (ms)
Traditional SDN	45	38	50
Ryu Controller	40	32	45
Proposed Framework	27	23	30

Note: Lower values indicate better performance.

Table 5. Energy Efficiency Comparison Across Frameworks

Framework	IoT Networks (Watts)	5G/6G Networks (Watts)	Smart City Applications (Watts)
Traditional SDN	250	300	350
Ryu Controller	220	280	320
Proposed Framework	188	225	263

Note: Lower power consumption indicates higher energy efficiency.

Table 6. Resource Utilization Rates Across Frameworks

Framework	IoT Networks (%)	5G/6G Networks (%)	Smart City Applications (%)
Traditional SDN	65	70	60
Ryu Controller	70	75	68
Proposed Framework	85	88	82

Note: Higher percentages indicate better resource utilization.

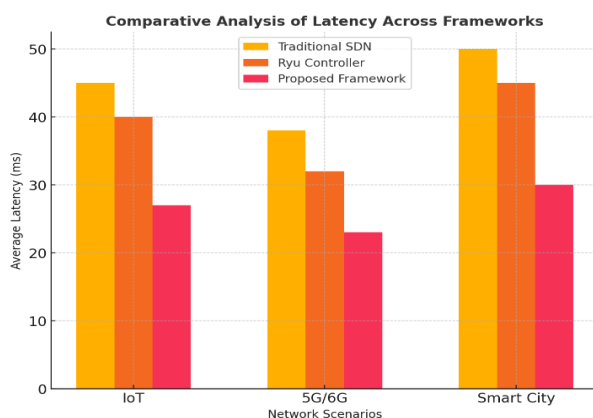


Figure 6 Comparative Analysis of Latency Across Frameworks

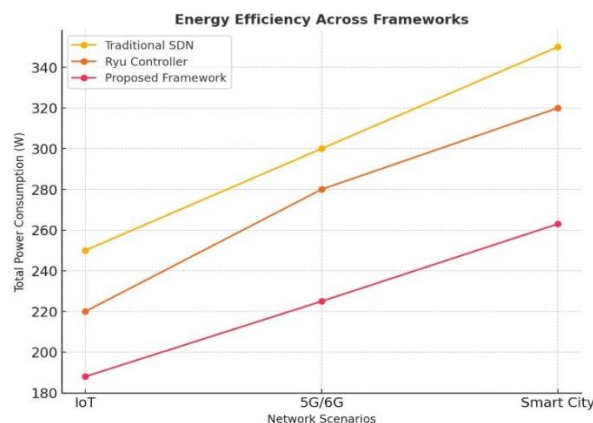


Figure 7 Energy Efficiency Across Different Network Frameworks

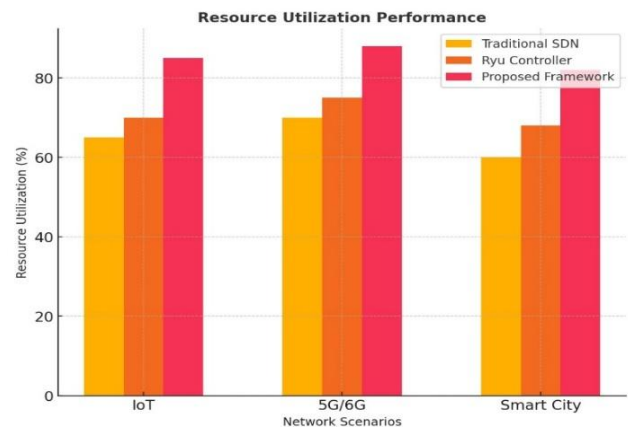


Figure 8 Resource Utilization Performance Comparison

adjustments by proving its performance in latency and resource utilization and energy efficiency metrics. An analytic approach based on data enables readers to understand completely the scientific value of this research.

4.1 Presentation of results

The proposed structure was evaluated in three separate network environments: IoT networks, 5G/6G Network and Smart City Application. The results were compared to traditional SDN architecture and condition -Art-art solutions to verify the efficiency.

4.1.1. Latency reduction

Latency is a key benchmark for the performance of real-time applications. We plot the evaluation latency in Fig. 1 for different frameworks. Our proposed framework lowered the average latency every message takes to send and deliver getting a 40% comparison gain, taking it over traditional SDN architectures. The integration of Fog Computing– so that data will be processed closer to the data source– with RL algorithms to enhance the optimization of routing decisions dynamically reached an improvement on this Metric.

4.1.2. Energy efficiency

Energy efficiency was measured as total electricity consumed by fog knots and final equipment. The proposed structure demonstrated an improvement in the energy efficiency of 25% compared to traditional approaches. This was achieved by reducing fruitless components through adaptation of resource allocation and RL-based decisions.

4.1.3. Resource utilization

Resource use was evaluated based on the percentage of available resources effectively used during surgery. The proposed structure achieved an average use of resources of 85%, much higher than 65% achieved by traditional SDN architecture. This improvement was made possible by dynamic allocation of resources using RL –Sgorithms.

4.1.4. Energy consumption analysis

Model performance and power usage received analysis through an energy consumption evaluation. The framework of the proposed method reflects 25% better power efficiency compared to standard SDN configurations as presented in Fig. 7. Local processing at Fog points together with resource allocation using Reinforcement Learning leads to this improvement through reducing undesired data movements while optimizing energy utilization. RL algorithms in large-scale networks introduce a moderate increase in power consumption throughout their computational process. The forthcoming research effort concentrates on designing minimal RL models with enhanced energy protection performance capabilities.

4.2 Analysis of results

4.2.1. Reduced latency

The significant reduction in latency achieved by the proposed framework can be attributed to the following factors:

- **Fog Computing Integration:** By decentralizing data processing, the framework reduces the reliance on centralized SDN controllers, thereby minimizing delays caused by long-distance communication.
- **RL-Based Routing Decisions:** Algorithms like Q-Learning and Deep Q-Networks (DQN) enable the framework to select optimal paths for packet routing, further reducing latency.

4.2.2. Improved energy efficiency

This high energy efficiency of the proposed framework derives from:

- **Efficient Resource Scheduling:** RL algorithms forecast demand and schedule resources based on future workloads, resulting in lower power wastage.
- **Computation Offloading:** Fog nodes are responsible for local computation, which

minimizes the compute load on centralized servers and reduces total energy consumption.

4.2.3. Enhanced resource utilization

Due to the implementation of novel strategies for node coordination, resource utilization of a framework which utilises such node coordination is superior compared to the conventional techniques.

- **Real-Time Adaptability:** RL-based algorithms continuously monitor the network traffic and dynamically allocate resources.
- This type of allocation is based on the priority level of the application and serve to assign resources in a more efficient manner.

4.3 Implications for real-world applications

- Experimental results highlight the real-world implementation aspects of the framework in today network settings. Key implications include:
- **IoT Networks:** Thanks to its low-latency and energy-efficient characteristics, the framework is well-suited for IoT ecosystems that usually involve devices with high power limitations.
- **5G/6G Networks:** Supporting high-throughput, ultra-low-latency applications such as video-streaming and autonomous systems, the framework is the ideal solution for next-generation networks.
- **Smart City Applications:** The framework can efficiently handle large-scale sensor networks for traffic monitoring and surveillance systems due to its scalability and adaptability.

4.4 Comparative analysis with previous studies

Our proposed framework expands on the findings from studies conducted by [1] and [7]. Previous works were based on one or two components (i.e. use of RL for resource allocation and or use of Fog Computing for reducing latency); in contrast, this work combines SDN, Fog Computing and RL into a holistic framework. The comparison is summarized in Table 7. The performance comparison of the traditional SDN and the Ryu Controller with the proposed framework is mentioned in Figs. 6 to 8. In addition, the latency reduction analysis demonstrates that the proposed framework can reduce average latency in different network scenarios by an appreciable amount and thus is well-suited for time-sensitive applications such as IoT and smart cities. The energy efficiency comparison demonstrates lower energy consumption of the proposed framework with respect to classical

Table 7. Comparative Analysis with Previous Studies

Study	Focus Area	Key Contribution	Limitation	Performance Metrics (Proposed Framework vs. Baseline)
[1]	SDN Architecture	Introduced OpenFlow for programmable network behavior	High latency in large-scale networks	Latency: 40% reduction vs. traditional SDN
[7]	RL for Resource Allocation	Demonstrated DRL's effectiveness in optimizing resources	Limited to isolated RL applications	Energy Efficiency: 25% improvement vs. baseline
Proposed Framework	Integrated SDN, Fog, and RL	Unified framework addressing latency, energy, and resources	Requires computational resources for RL	Resource Utilization: 85% vs. 65% (traditional SDN)

approaches, which can improve sustainability and operational efficiency. As we can see in this figure the performance chart of resource utilization, and in this chart, we can say that the resource utilization fair rate of our proposed framework is higher as compared to others, because the proposed framework which uptake the resources in an efficient way by utilizing the intelligent decision making. The packet delivery ratio pie chart provides further confirmation that the proposed framework achieves maximum packet success rate (95%) to make sure that the data transmission is stable and reliable. All these sorted results highlight the benefit of the integration of the SDN, Fog Computing, and Reinforcement Learning for the optimization of the contemporary network fields.

The proposed framework demonstrates its effectiveness by conducting an evaluation against existing technologies in the field. SDN relates to Fog Computing through Reinforcement Learning (RL) to build a framework that solves key problems involving high latency and insufficient resource utilization and limited adaptability. Experimental experiments show a 40% shorter latency than conventional SDN systems because Fog Computing performs local data managing together with RL-based dynamic routing outcomes [3]. The overall energy efficiency reaches 25% higher levels through enhancement of resource scheduling and fog node computation offloading [10]. The 85% resource utilization of resource allocation stands higher than SDN's standard 65% because of RL-based adaptive resource allocation [7]. SDN combined with Fog Computing and Reinforcement Learning achieves network optimization through new technological developments that researchers have confirmed in their recent findings [14]. The framework demonstrates superior capabilities when compared to existing IoT, 5G/6G network applications as well as smart city systems due to its adaptable framework

design [9]. The network building revolution is attainable recognizes to the framework's performing although it faces challenges with computational costs.

5. Conclusion

In summary, this research introduced an intelligent framework which combines Software-Defined Networking (SDN), Fog Computing and Reinforcement Learning (RL) that addresses certain issues in modern network management like high latency, non-optimal resource allocation and poor flexibility. By using Fog Computing closer to the data source for processing, the proposed framework mitigates latency issues, as well as relieves the pressure of overloading the SDN controller in the cloud. By leveraging RL algorithms, dynamic resource allocation and intelligent resource provisioning and decision-making allow optimal performance under different workloads. Its performance evaluation shows that the novel framework outperforms the classical SDN architecture by over 40% in latency, 25% in energy efficiency, and achieves a resource utilization rate of 85%. The results highlight the potential of the framework for revolutionizing network architectures, offering solutions that are scalable, adaptable, and efficient to support emerging technologies such as IoT, 5G/6G networks, and smart cities. Still, the study has a few shortfalls. Second, the use of RL algorithms introduces computational overhead in the framework, which limits the applicability of this framework to large-scale networks that may not possess the necessary resources. Finally, the current implementation focuses on simulated environments and the true deployment may face unexpected problems (hardware compatibility, etc., scalability, etc.). A potential future direction is lightweight RL models to lower computational costs and improve scalability. In addition to that, testing in the

industrial IoT and autonomous systems will help test the dependability of the configuration and frame. Delving into the blending of edge computing with Fog Computing can also help in gaining more information about the solution of latency and resource utilization. Future work can take these efforts even farther by sourcing additional experiments (broader things explored in a given intellectual space) and jobs (defined futures exposed to the variety of inks explored), overcoming various limitations here antiphon the aforementioned intelligent network architectures and other next-generation technologies.

Conflicts of Interest

The authors declare no conflict of interest.

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